WORKED SOLUTION

STEP I 2016, Question 5

Pure • Touching Circles

STEP I 2016, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q5

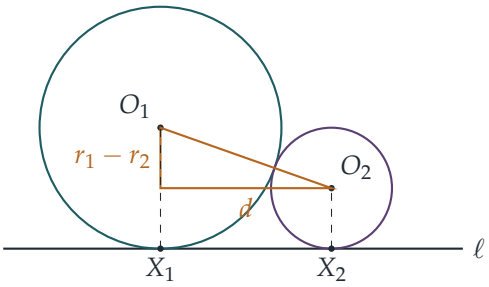
The key idea

Everything rests on one fact: if two circles touch each other and both touch the same straight line, the distance between their points of contact with the line is $2\sqrt{r_1 r_2}$. That comes from a single right-angled triangle. After that, the substitution $x = 1/\sqrt{a}$, $y = 1/\sqrt{b}$, $z = 1/\sqrt{c}$ turns $(**)$ into a symmetric quartic which factorises into four linear factors, and one of those factors is exactly $(*)$.

Solution

Part (i)

Two touching circles on a line. Take two circles of radii r_1 and r_2 touching each other externally, both on the same side of a line ℓ and both tangent to it, at points X_1 and X_2. The radii to X_1 and X_2 are perpendicular to ℓ, so the centres O_1, O_2 lie at heights r_1 and r_2 above ℓ. Since the circles touch externally, $O_1O_2 = r_1 + r_2$.



Drop the perpendicular from O_2 to the radius O_1X_1 (shown with $r_1 \geq r_2$). This gives a right-angled triangle with hypotenuse $r_1 + r_2$, vertical side $|r_1 - r_2|$ and horizontal side $d = X_1X_2$. By Pythagoras,

$$d^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2 = 4r_1r_2, \qquad \text{so} \qquad X_1X_2 = 2\sqrt{r_1 r_2}.$$

Applying it three times. In the diagram each pair of circles touches and all three touch the line at P, Q, R, with Q between P and R. So

$$PQ = 2\sqrt{ab}, \qquad QR = 2\sqrt{bc}, \qquad PR = 2\sqrt{ac},$$

and $PQ + QR = PR$ gives $\sqrt{ab} + \sqrt{bc} = \sqrt{ac}$. Dividing by $\sqrt{abc} > 0$,

$$\frac{1}{\sqrt{c}} + \frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}},$$

which is $(*)$.

Deducing $(**)$. Write $x = \dfrac{1}{\sqrt{a}}$, $y = \dfrac{1}{\sqrt{b}}$, $z = \dfrac{1}{\sqrt{c}}$, so that $\dfrac{1}{a} = x^2$ and so on, and $(*)$ says $y = x + z$. Then $(**)$ reads $2(x^4 + y^4 + z^4) = (x^2 + y^2 + z^2)^2$; expanding the right-hand side and rearranging, it is equivalent to

$$F := x^4 + y^4 + z^4 - 2x^2y^2 - 2y^2z^2 - 2z^2x^2 = 0.$$

? **Ask yourself** How can this quartic be factorised?

Group it as a difference of two squares. Since $(x^2 + y^2 - z^2)^2 = x^4 + y^4 + z^4 + 2x^2y^2 - 2y^2z^2 - 2z^2x^2$,

$$\begin{aligned} F &= (x^2 + y^2 - z^2)^2 - (2xy)^2 \\ &= (x^2 - 2xy + y^2 - z^2)(x^2 + 2xy + y^2 - z^2) \\ &= ((x-y)^2 - z^2)((x+y)^2 - z^2) \\ &= (x - y - z)(x - y + z)(x + y - z)(x + y + z). \end{aligned}$$

By $(*)$, $x - y + z = 0$, so $F = 0$ and $(**)$ holds.

Part (ii)

Now a, b, c are any positive numbers with $b < c < a$ satisfying $(**)$. Define x, y, z as before; they are positive, and because $t \mapsto 1/\sqrt{t}$ is decreasing on $t > 0$, the condition $b < c < a$ gives

$$y > z > x > 0.$$

As shown in part (i), $(**)$ is equivalent to

$$(x - y - z)(x - y + z)(x + y - z)(x + y + z) = 0,$$

so one of the four factors vanishes. Three of them cannot:

- $x + y + z > 0$, since all three are positive;
- $x + y - z > 0$, since $y > z$ and $x > 0$;
- $x - y - z < 0$, since $x < y$ and $z > 0$.

Hence $x - y + z = 0$, that is $y = x + z$, or

$$\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}},$$

which is $(*)$. (Only the fact that b is the smallest of the three was needed.)

Checks

- Take $a = 4$, $c = 1$. Then $(*)$ gives $1/\sqrt{b} = \frac{1}{2} + 1 = \frac{3}{2}$, so $b = \frac{4}{9}$. Now $2\left(\frac{1}{16} + \frac{81}{16} + 1\right) = \frac{49}{4}$ and $\left(\frac{1}{4} + \frac{9}{4} + 1\right)^2 = \left(\frac{7}{2}\right)^2 = \frac{49}{4}$, as $(**)$ requires.
- Why is the ordering needed in (ii)? With $a = 4$, $c = 1$, the relation $(**)$ is a quadratic in $1/b$ with two roots: $b = \frac{4}{9}$ and $b = 4$ (from $1/\sqrt{b} = |x - z| = \frac{1}{2}$). The second has b largest rather than smallest, and it satisfies $(**)$ but not $(*)$: here it is a and b that are tied, through $1/\sqrt{c} = 1/\sqrt{a} + 1/\sqrt{b}$. Swapping the roles of the letters corresponds to swapping which circle sits in the middle.
- A computer check on thousands of random configurations of three touching circles on a line confirmed both $(*)$ and $(**)$.

Common slips

- Using the distance between the centres, $r_1 + r_2$, as the distance between the points of contact: the centres are at different heights.
- In part (ii), concluding $(*)$ from the factorisation without ruling out the other three factors; this is exactly where $b < c < a$ is used.

Beyond STEP

Relation $(**)$ is the degenerate case of Descartes' circle theorem: for four mutually touching circles with curvatures $k_i = 1/r_i$, one has $2\sum k_i^2 = \left(\sum k_i\right)^2$, and a straight line counts as a circle of curvature 0. Repeatedly fitting new circles into the gaps produces the Apollonian gasket, where integer curvatures lead to surprising number theory. The factorisation of F is also the one behind Heron's formula, $16\Delta^2 = -F$ with x, y, z the sides of a triangle. For more on circle geometry and inversion, which makes these results transparent, see Needham, *Visual Complex Analysis*.

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